

A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.4

(2304102L)

Submitted by,

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4.1.1. Pandas - series creation and manipulation

03:12

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd
```

```
# # Take inputs from the user to create a list of numbers
```

```
# numbers = list(map(int, input().split()))
```

```
# grouped =
```

```
# # Display the mean of even and odd numbers with labels #
```

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in
```

```
grouped.index] # print("Mean of even and odd numbers:")
```

```
# print(grouped) import
```

```
pandas as pd
```

```
# Take inputs from the user to create a list of numbers numbers
```

```
= list(map(int, input().split()))
```

```
# Create a Pandas series from the list of numbers
```

```
s = pd.Series(numbers)
```

```
# Grouping by even and odd numbers and calculating the mean grouped
```

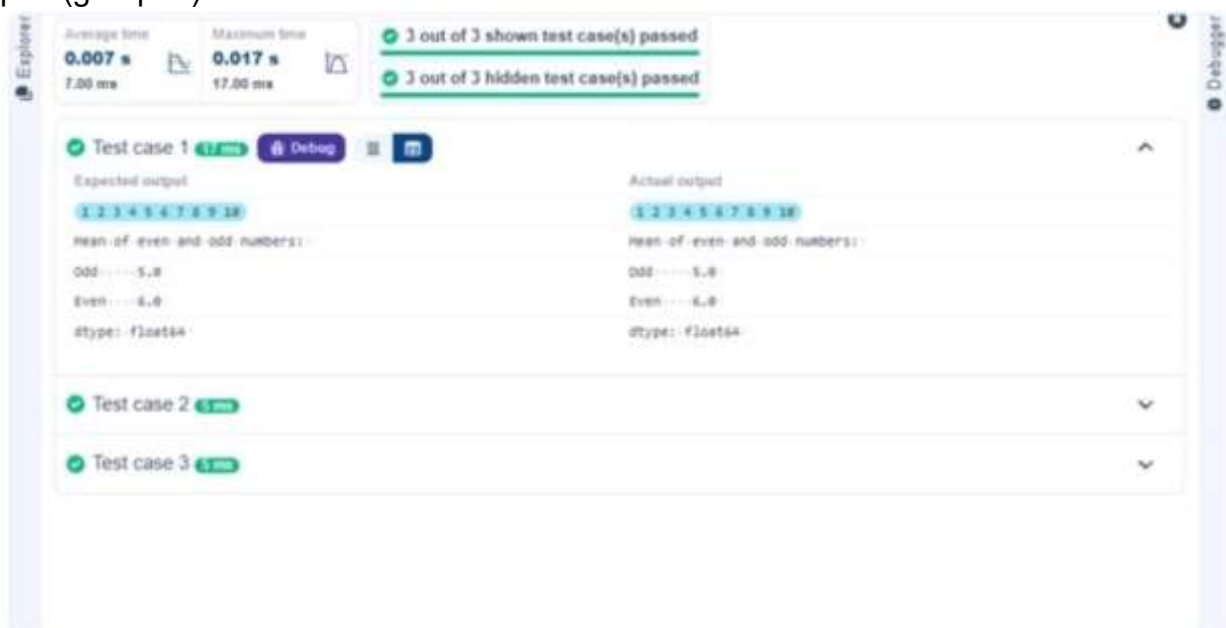
```
= s.groupby(s % 2 == 0).mean()
```

```
# Display the mean of even and odd numbers with labels
```

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
print("Mean of even and odd numbers:")
```

```
print(grouped)
```



4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).

Add the new row to the DataFrame.

Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.

Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.

Remove the specified row.

Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.

Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.

Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

Display the DataFrame after deleting the column.

```
import pandas as pd
# Provided dictionary of lists data
= {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
```

```
}
```

```
# Convert the dictionary to a
```

```
DataFrame df = pd.DataFrame(data)
```

```
# Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Adding a new row new_name =
```

```
input("New name: ") new_age =
```

```
int(input("New age: ")) df.loc[len(df)] =
```

```
[new_name, new_age]
```

```
# Display the DataFrame after adding a new row print("After  
adding a row:\n",df)
```

```
# Modifying a row row_to_modify = int(input("Index  
of row to modify: ")) new_age_value = int(input("New  
age: ")) df.at[row_to_modify, "Age"] =  
new_age_value
```

```
# Display the DataFrame after modifying a row print("After  
modifying a row:")
```

```
print(df)
```

```
# Deleting a row row_to_delete = int(input("Index of  
row to delete: "))  
df = df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row print("After  
deleting a row:")  
print(df)
```

```
# Adding a new column genders_input = input("Enter  
genders separated by space: ") genders =  
genders_input.split()
```

```
# Ensure correct length if  
len(genders) == len(df):  
    df["Gender"] = genders  
else:  
    print("Error: Number of genders must match rows!") df["Gender"]  
    = ["Unknown"] * len(df)
```

```
# Display the DataFrame after adding a new column  
print("After adding a new column:")
```

```
print(df)
```

```
# Modifying a column
```

```
df["Name"] = df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column print("After  
modifying a column:")
```

```
print(df)
```

```
# Deleting a column
```

```
df = df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column print("After  
deleting a column:")
```

```
print(df)
```


Average time
0.043 s
43.00 ms



Maximum time
0.049 s
49.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 **49 ms** **Debug**

Expected output

Original DataFrame:

..... Name .. Age

0 ... Alice ... 25

1 ... Bob ... 30

2 ... Charlie ... 35

New name: **Susan**

New age: **29**

After adding a row:

..... Name .. Age

0 ... Alice ... 25

1 ... Bob ... 30

2 ... Charlie ... 35

3 ... Susan ... 29

Index of row to modify: **4**

New age: **27**

After modifying a row:

..... Name .. Age

0 ... Alice ... 27

1 ... Bob ... 30

2 ... Charlie ... 35

3 ... Susan ... 29

Index of row to delete: **3**

After deleting a row:

..... Name .. Age

0 ... Alice ... 27

1 ... Bob ... 30

Actual output

Original DataFrame:

..... Name .. Age

0 ... Alice ... 25

1 ... Bob ... 30

2 ... Charlie ... 35

New name: **Susan**

New age: **29**

After adding a row:

..... Name .. Age

0 ... Alice ... 25

1 ... Bob ... 30

2 ... Charlie ... 35

3 ... Susan ... 29

Index of row to modify: **4**

New age: **27**

After modifying a row:

..... Name .. Age

0 ... Alice ... 27

1 ... Bob ... 30

2 ... Charlie ... 35

3 ... Susan ... 29

Index of row to delete: **3**

After deleting a row:

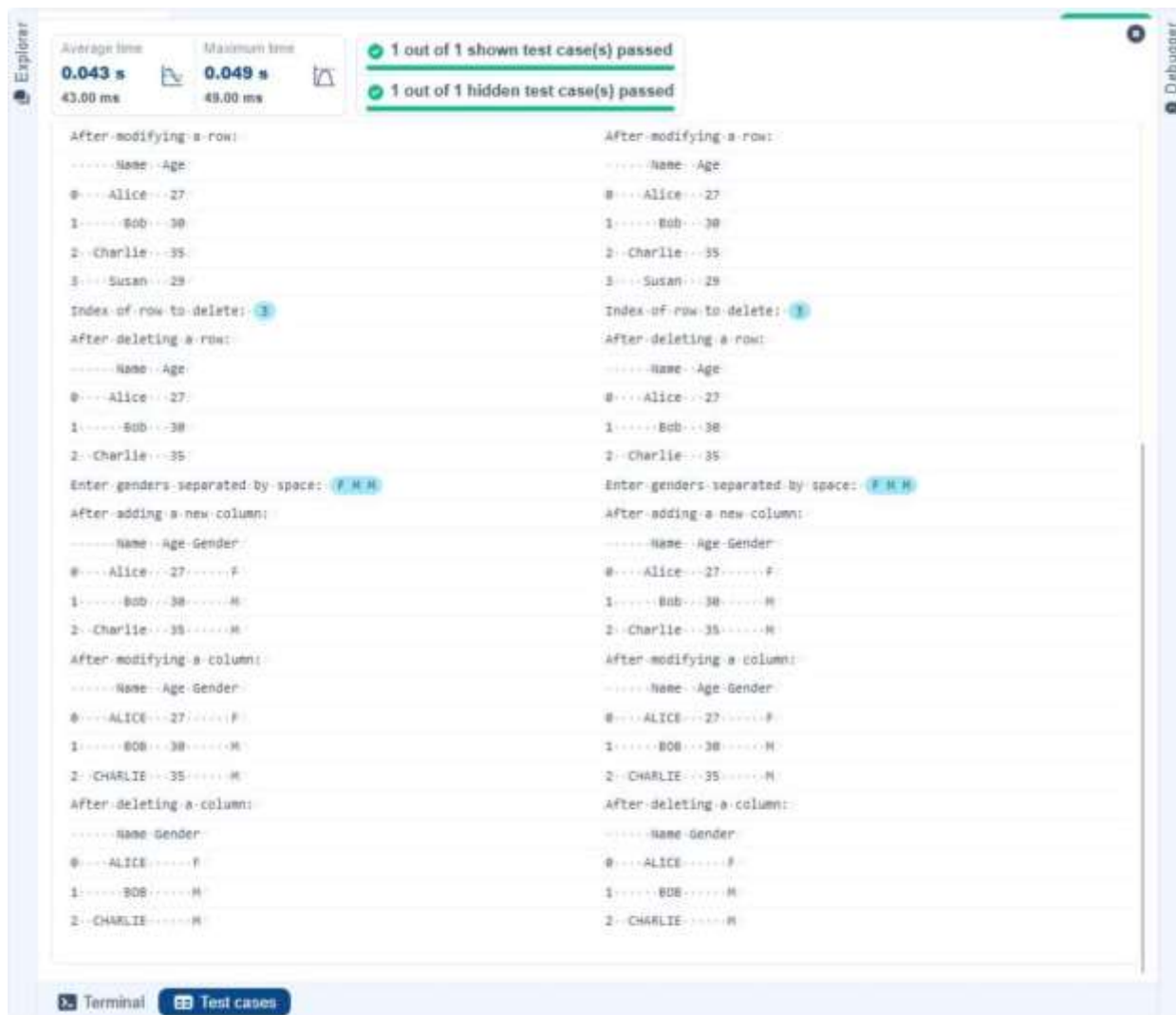
..... Name .. Age

0 ... Alice ... 27

1 ... Bob ... 30

Terminal

Test cases



4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd

# Read the text file into a DataFrame
file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age",
"Grade"]) # Display the first five rows
print("First five rows:")
print(data.head())

# Calculate the average age (2 decimal places)
avg_age = round(data["Age"].mean(), 2)
print("Average age:", avg_age)

# Filter students with grade up to 'B' (A and B) filtered
= data[data["Grade"].isin(['A', 'B'])]
```

```
print("Students with a grade up to B")
```

```
print(filtered)
```

The screenshot displays a coding platform interface with the following components:

- Performance Metrics:**
 - Average time: 0.028 s (27.50 ms)
 - Maximum time: 0.040 s (40.00 ms)
- Test Case Status:**
 - 1 out of 1 shown test case(s) passed
 - 1 out of 1 hidden test case(s) passed
- Test Case 1 Details:**
 - Expected output:**

```
studentdata.txt
First five rows:
-- Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
3 Mike 23 C
4 Sony 21 B
Average age: 23.25
Students with a grade up to B:
-- Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
4 Sony 21 B
5 Ania 26 A
```
 - Actual output:** (Identical to the expected output)
- Navigation:** Buttons for Terminal, Test cases, Prev, Reset, Submit, and Next.

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd

# Prompt the user for the file name file_name
file_name = input()

# Load the data df =
df = pd.read_csv(file_name) # Prepare
data for calculation df['Date'] =
pd.to_datetime(df['Date']) df['Month']
= df['Date'].dt.to_period('M')
df['Sales'] = df['Quantity'] * df['Price']

# Find the month with the highest total sales best_month
best_month = df.groupby('Month')['Sales'].sum().idxmax()
highest_sales = df.groupby('Month')['Sales'].sum().max()
```

```
print(f"Best month: {best_month}") print(f"Total
sales: ${highest_sales:.2f}")
```



4.2.2. Best Selling

Product 02:08

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New

York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los

Angeles

2025-01-05,Product **A**,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
# Find the product with the highest total quantity sold
```

```
best_product =
```

```
df.groupby('Product')['Quantity'].sum().idxmax()
```

```
highest_quantity =
```

```
df.groupby('Product')['Quantity'].sum().max() # Display the
```

```
result print(f"Best selling product: {best_product}")
```

```
print(f"Total quantity sold: {highest_quantity}")
```



4.2.3. City that Sold the Most

Products 01:45

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by City and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New

York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los

Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
# Group by City and calculate total quantity city_sales
```

```
= df.groupby('City')['Quantity'].sum()
```

```
# Find the city with the highest total quantity sold best_city
```

```
= city_sales.idxmax()
```

```
# Display the result print(f"City sold the most
```

```
products: {best_city}")
```



4.2.4. Most Frequently Sold Product

Pairs 01:25

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product **A**,5,20,New

York

2025-01-01,Product **B**,3,15,Los Angeles

2025-01-02,Product **A**,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product **A**,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los

Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B : Appear in transactions on 2025-01-01 and 2025-01-03.

Product A and Product C : Appear in transactions on 2025-01-02 and 2025-01-05.

Product B and Product C : Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

Product A and Product B (2 times)

Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd

from itertools import combinations from
collections import Counter

# Prompt user to input the file name file_name
= input()

# Read data from the specified CSV file
df = pd.read_csv(file_name) # write the
code pair_counter = Counter()

# Group by Date and generate product pairs
for _, group in df.groupby('Date'): products =
group['Product'].tolist() pairs =
combinations(sorted(products),
    2) pair_counter.update(pairs)

# Find maximum frequency
max_count = max(pair_counter.values())

# Output the most frequent product pairs
for pair, count in pair_counter.items(): if
count == max_count:
```

```
print(f"{pair[0]} and {pair[1]}: {count} times")
```

Average time

0.020 s

20.33 ms

Maximum time

0.039 s

39.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 30 ms Debug Test cases

Expected output

`sales_data.csv`

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

`sales_data.csv`

Product A and Product B: 2 times

Product A and Product C: 2 times

Terminal

Test cases

4.2.5. Titanic Dataset Analysis and Data

Cleaning 01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
```

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset
```

```
data =
```

```
pd.read_csv('Titanic-Dataset.csv') # 1.
```

```
Display the first 5 rows of the dataset
```

```
print(data.head())
```

```
# 2. Display the last 5 rows of the
```

```
dataset print(data.tail())
```

```
# 3. Get the shape of the dataset print(data.shape)
```

```
# 4. Get a summary of the dataset (info)
```

```
print(data.info())
```

5. Get basic statistics of the dataset `print(data.describe())`

6. Check for missing values

`print(data.isnull().sum())`

7. Fill missing values in the 'Age' column with the median age `data['Age'] = data['Age'].fillna(data['Age'].median())`


```
# 8. Fill missing values in the 'Embarked' column with the mode data['Embarked']
= data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
# 9. Drop the 'Cabin' column due to many missing values
data = data.drop(columns=['Cabin'])
```

```
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch' data['FamilySize']
= data['SibSp'] + data['Parch']
```

Explorer

Average time

0.166 s

166.00 ms

Maximum time

0.166 s

166.00 ms

1 out of 1 shown test case(s) passed

Test case 1

166 ms

Debug

Expected output

Actual output

PassengerId

Survived

Pclass

Fare

Cabin

Embarked

0

1

0

3

7.2500

NaN

S

1

2

1

1

71.2833

C85

C

2

3

1

3

7.9250

NaN

S

3

4

1

1

53.1000

C123

S

4

5

0

3

8.0500

NaN

S

PassengerId

Survived

Pclass

Fare

Cabin

Embarked

0

1

0

3

7.2500

NaN

S

1

2

1

1

71.2833

C85

C

2

3

1

3

7.9250

NaN

S

3

4

1

1

53.1000

C123

S

4

5

0

3

8.0500

NaN

S

[5 rows x 12 columns]

[5 rows x 12 columns]

PassengerId

Survived

Pclass

Fare

Cabin

Embarked

886

887

0

2

13.00

NaN

S

887

888

1

1

30.00

B42

S

888

889

0

3

23.45

NaN

S

889

890

1

1

30.00

C140

C

890

891

0

3

7.75

NaN

Q

PassengerId

Survived

Pclass

Fare

Cabin

Embarked

886

887

0

2

13.00

NaN

S

887

888

1

1

30.00

B42

S

888

889

0

3

23.45

NaN

S

889

890

1

1

30.00

C140

C

890

891

0

3

7.75

NaN

Q

Average time

0.166 s
166.00 ms

Maximum time

0.166 s
166.00 ms

1 out of 1 shown test case(s) passed

[5 rows x 12 columns]

(891, 12)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

Column Non-Null Count Dtype

0 PassengerId 891 non-null int64

1 Survived 891 non-null int64

2 Pclass 891 non-null int64

3 Name 891 non-null object

4 Sex 891 non-null object

5 Age 714 non-null float64

6 SibSp 891 non-null int64

7 Parch 891 non-null int64

8 Ticket 891 non-null object

9 Fare 891 non-null float64

10 Cabin 204 non-null object

11 Embarked 891 non-null object

dtypes: float64(2), int64(5), object(5)

memory usage: 66.2+ KB

None

PassengerId Survived Pclass SibSp Parch

arch Fare

count 891.000000 891.000000 891.000000 891.000000 891.00

mean 446.000000 0.343750 2.308642 0.523086 0.30

std 257.353842 0.476592 0.836871 1.302743 0.80

min 0.000000 0.000000 1.000000 0.000000 0.00

max 891.000000 1.000000 3.000000 8.000000 512.00

[5 rows x 12 columns]

(891, 12)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

Column Non-Null Count Dtype

0 PassengerId 891 non-null int64

1 Survived 891 non-null int64

2 Pclass 891 non-null int64

3 Name 891 non-null object

4 Sex 891 non-null object

5 Age 714 non-null float64

6 SibSp 891 non-null int64

7 Parch 891 non-null int64

8 Ticket 891 non-null object

9 Fare 891 non-null float64

10 Cabin 204 non-null object

11 Embarked 891 non-null object

dtypes: float64(2), int64(5), object(5)

memory usage: 66.2+ KB

None

PassengerId Survived Pclass SibSp Parch

arch Fare

count 891.000000 891.000000 891.000000 891.000000 891.00

mean 446.000000 0.343750 2.308642 0.523086 0.30

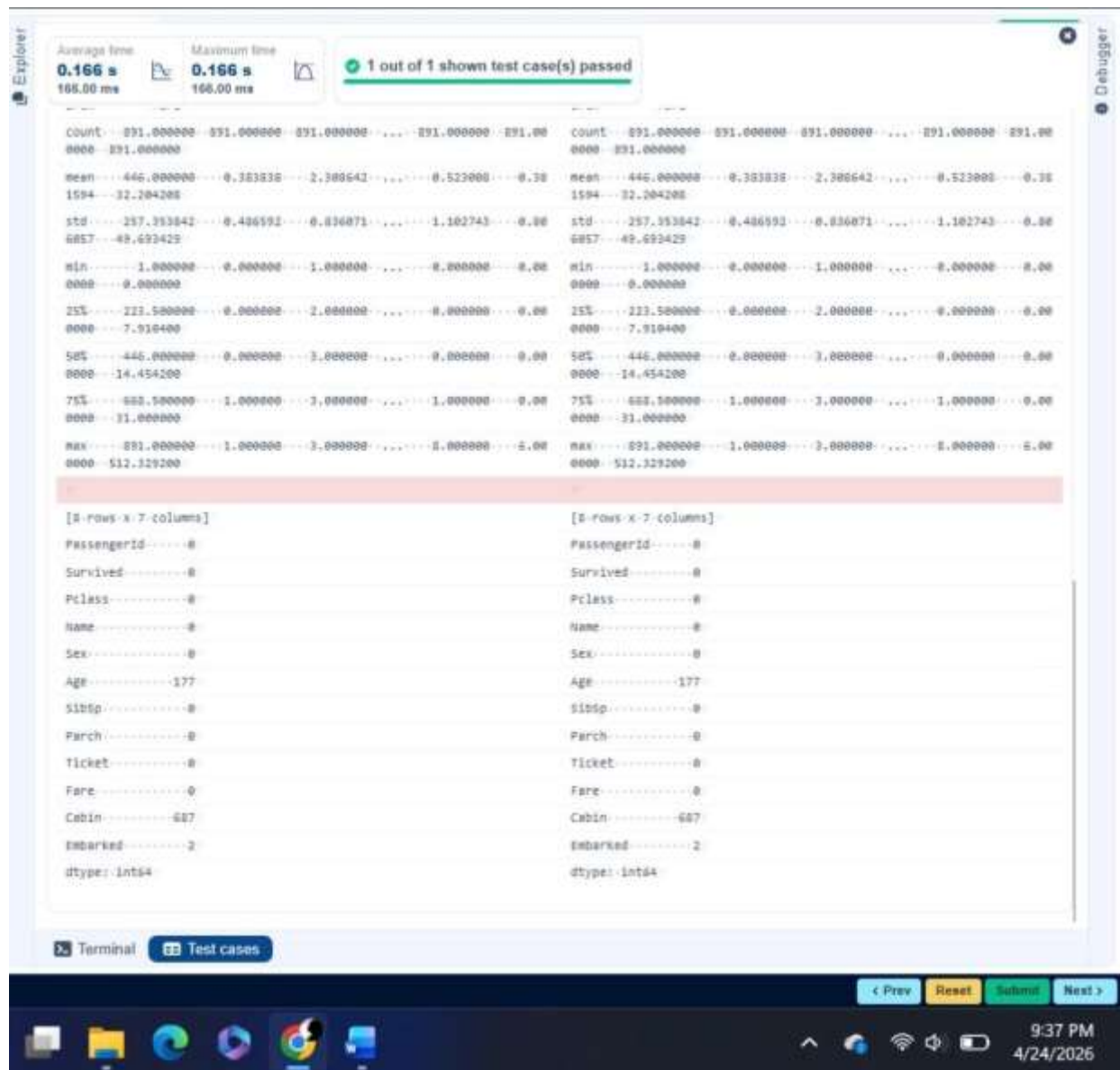
std 257.353842 0.476592 0.836871 1.302743 0.80

min 0.000000 0.000000 1.000000 0.000000 0.00

max 891.000000 1.000000 3.000000 8.000000 512.00

Terminal

Test cases



4.2.6. Titanic Dataset Analysis and Data Cleaning -

2 02:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.

7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embar k

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-  
Dataset.csv') data['FamilySize'] = data['SibSp'] +  
data['Parch'] # 1. Create a new column 'IsAlone' (1 if alone,  
0 otherwise) data['IsAlone'] = np.where(data['FamilySize']  
== 0, 1, 0)
```

```
# 2. Convert 'Sex' to numeric (male: 0, female: 1) data['Sex']  
= data['Sex'].map({'male': 0, 'female': 1})
```

```
# 3. One-hot encode the 'Embarked' column  
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 4. Get the mean age of passengers  
print(data['Age'].mean())
```

```
# 5. Get the median fare of passengers  
print(data['Fare'].median())
```

```
# 6. Get the number of passengers by class print(data['Pclass'].value_counts())
```

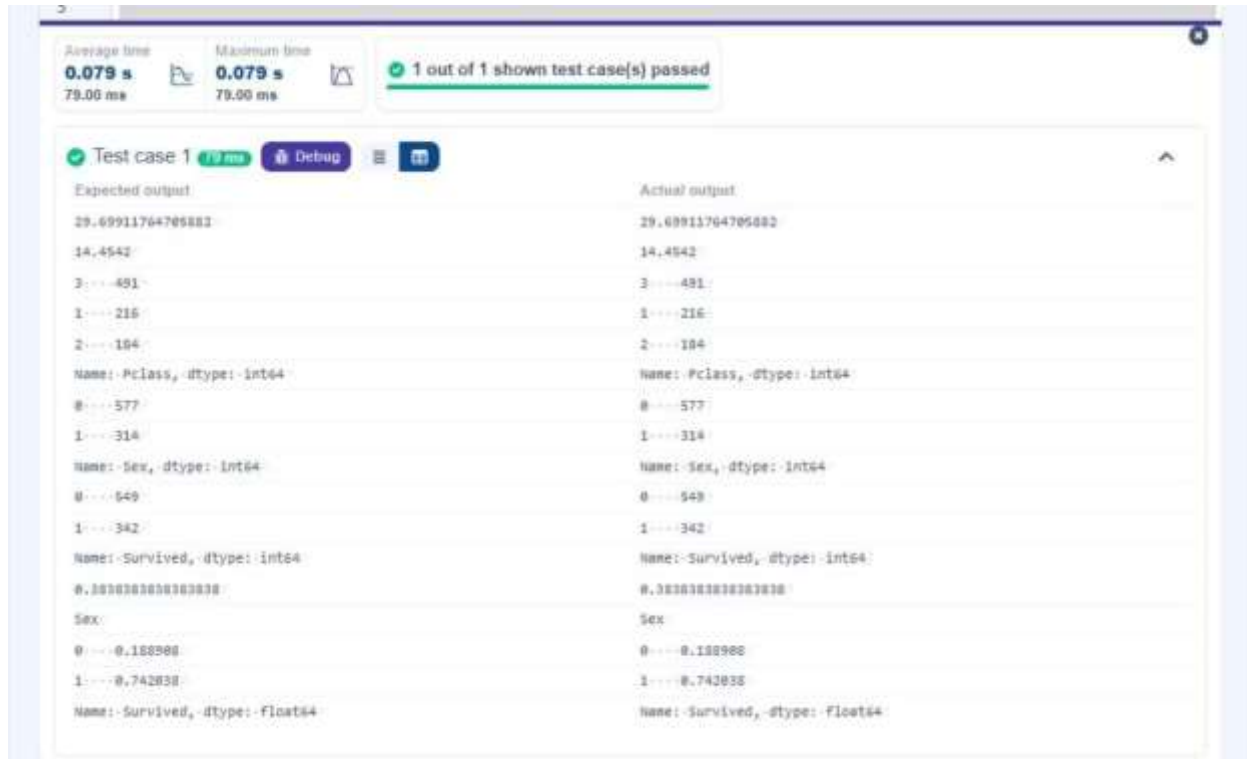
```
# 7. Get the number of passengers by gender print(data['Sex'].value_counts())
```

```
# 8. Get the number of passengers by survival status  
print(data['Survived'].value_counts())
```

9. Calculate the survival rate

```
print(data['Survived'].mean())
```

10. Calculate the survival rate by gender `print(data.groupby('Sex')['Survived'].mean())`



4.2.7. Titanic Dataset Analysis and Data Cleaning -

3 03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).

8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset data = pd.read_csv('Titanic-
Dataset.csv') data['FamilySize'] = data['SibSp'] +
data['Parch'] data['IsAlone'] =
np.where(data['FamilySize'] > 0, 0, 1) data =
pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # 1. Calculate the survival rate by
class print(data.groupby('Pclass')['Survived'].mean())
```


2. Calculate the survival rate by embarked location

```
print(data.groupby('Embarked_S')['Survived'].mean())
```

3. Calculate the survival rate by family size

```
print(data.groupby('FamilySize')['Survived'].mean())
```

4. Calculate the survival rate by being alone

```
print(data.groupby('IsAlone')['Survived'].mean())
```

5. Get the average fare by class

```
print(data.groupby('Pclass')['Fare'].mean())
```

6. Get the average age by class `print(data.groupby('Pclass')['Age'].mean())`

7. Get the average age by survival status

```
print(data.groupby('Survived')['Age'].mean()) # 8. Get the average fare by survival
```

```
status print(data.groupby('Survived')['Fare'].mean())
```

9. Get the number of survivors by class `print(data[data['Survived']`

```
==
```

```
1]['Pclass'].value_counts())
```

10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Explore

Average time: 0.103 s
Maximum time: 0.103 s
1 out of 1 shown test case(s) passed

Test case 1 100 ms Debug

Expected output	Actual output
Pclass	Pclass
1 --- 0.629638	1 --- 0.629638
2 --- 0.472826	2 --- 0.472826
3 --- 0.242382	3 --- 0.242382
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 --- 0.388873	0 --- 0.388873
1 --- 0.338953	1 --- 0.338953
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 --- 0.383538	0 --- 0.383538
1 --- 0.552795	1 --- 0.552795
2 --- 0.578411	2 --- 0.578411
3 --- 0.724138	3 --- 0.724138
4 --- 0.288888	4 --- 0.288888
5 --- 0.138364	5 --- 0.138364
6 --- 0.333333	6 --- 0.333333
7 --- 0.000000	7 --- 0.000000
10 --- 0.000000	10 --- 0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0 --- 0.585658	0 --- 0.585658
1 --- 0.383538	1 --- 0.383538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1 --- 84.154687	1 --- 84.154687
2 --- 20.662183	2 --- 20.662183
3 --- 11.675558	3 --- 11.675558
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1 --- 38.233441	1 --- 38.233441
2 --- 29.877638	2 --- 29.877638

Debugger

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

5 --- 0.136304
6 --- 0.133333
7 --- 0.000000
10 --- 0.000000

Name: Survived, dtype: float64

IsAlone

0 --- 0.505030
1 --- 0.303538

Name: Survived, dtype: float64

Pclass

1 --- 84.154687
2 --- 20.662183
3 --- 13.675558

Name: Fare, dtype: float64

Pclass

1 --- 38.233443
2 --- 29.877638
3 --- 25.188628

Name: Age, dtype: float64

Survived

0 --- 30.626179
1 --- 28.343000

Name: Age, dtype: float64

Survived

0 --- 22.117887
1 --- 48.395488

Name: Fare, dtype: float64

1 --- 138
3 --- 119
2 --- 87

Name: Pclass, dtype: int64

3 --- 172
2 --- 97
1 --- 80

5 --- 0.136304
6 --- 0.233333
7 --- 0.000000
10 --- 0.000000

Name: Survived, dtype: float64

IsAlone

0 --- 0.505030
1 --- 0.303538

Name: Survived, dtype: float64

Pclass

1 --- 84.154687
2 --- 20.662183
3 --- 13.675558

Name: Fare, dtype: float64

Pclass

1 --- 38.233443
2 --- 29.877638
3 --- 25.188628

Name: Age, dtype: float64

Survived

0 --- 30.626179
1 --- 28.343000

Name: Age, dtype: float64

Survived

0 --- 22.117887
1 --- 48.395488

Name: Fare, dtype: float64

1 --- 138
3 --- 119
2 --- 87

Name: Pclass, dtype: int64

3 --- 172
2 --- 97
1 --- 80

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